

Temporal Dependency of DLMs + Summary of Literature Review

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Temporal Dependency of Diffusion Models

- Activation distributions change with time so static quantization will lead to either large truncation error or rounding error

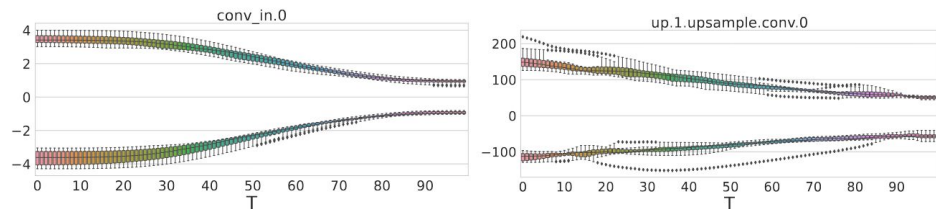


Figure 2: **Evolution of Activation Distribution of Diffusion Model over Time Steps.** Boxplot depicting the maximum and minimum values of activation for the DDIM model trained on the CIFAR-10 dataset across different time steps.

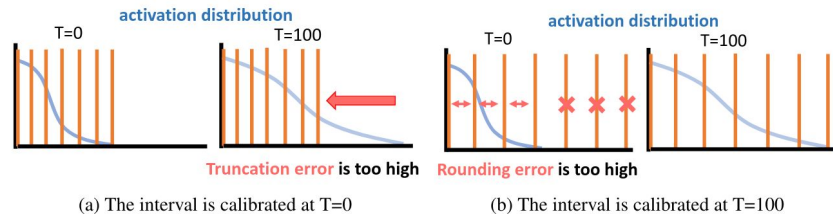


Figure 3: **Limitation of Static Quantization for Diffusion Models.** Assume that the activation distribution enlarges as time step progresses. (a) Large truncation error due to small quantization interval and (b) Large rounding error due to large quantization interval.

Temporal Dependency of Diffusion Models

- Measured the Pearson correlation between time steps and activation variation of the model showing that 62.1% of the layers exhibited moderate temporal dependence ($|r| > 0.5$), and 38.8% exhibited strong temporal dependence ($|r| > 0.7$)
- Use temporal dynamic quantization (TDQ), a trainable neural network that generates quantization interval using only timestep information

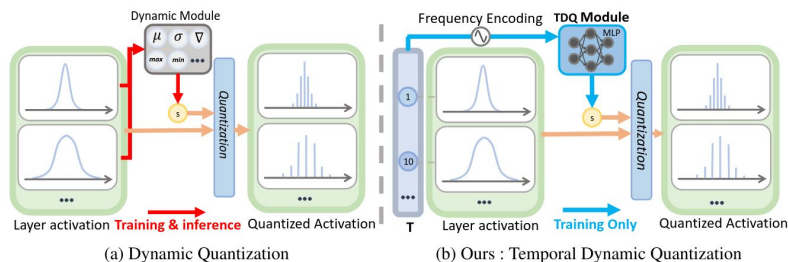


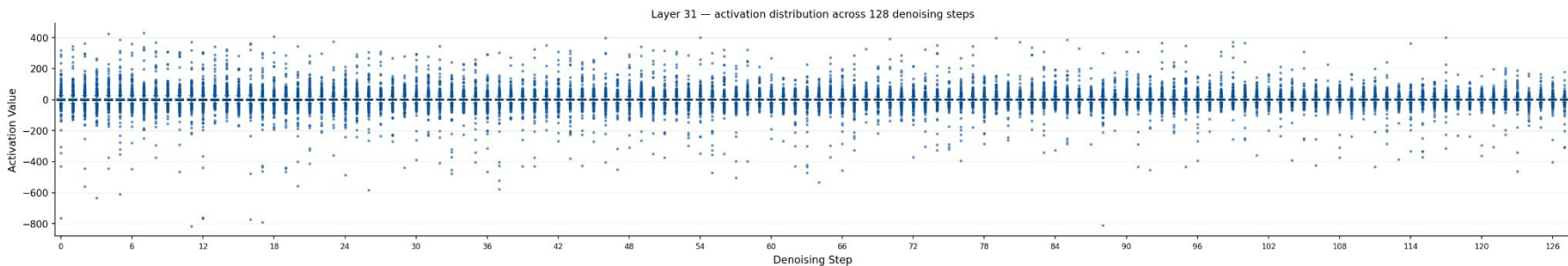
Figure 4: **Overview of TDQ module.** Comparison between (a) Input-dependent dynamic quantization, which requires activation statistics during both training and inference, and (b) the proposed TDQ module that enables cost-free inference based on pre-computed intervals.

Layer	Dim
Linear	128 x 64
ReLU	-
Linear	64 x 64
ReLU	-
Linear	64 x 64
ReLU	-
Dropout	(0.2)
Linear	64 x 1
Softplus	-

Table S3: Overview of TDQ Module architecture

Temporal Dependency of DLMs

- Variations in activation distributions appear across different timesteps in DLMs, especially in the outliers, these are averaged from 5 prompts

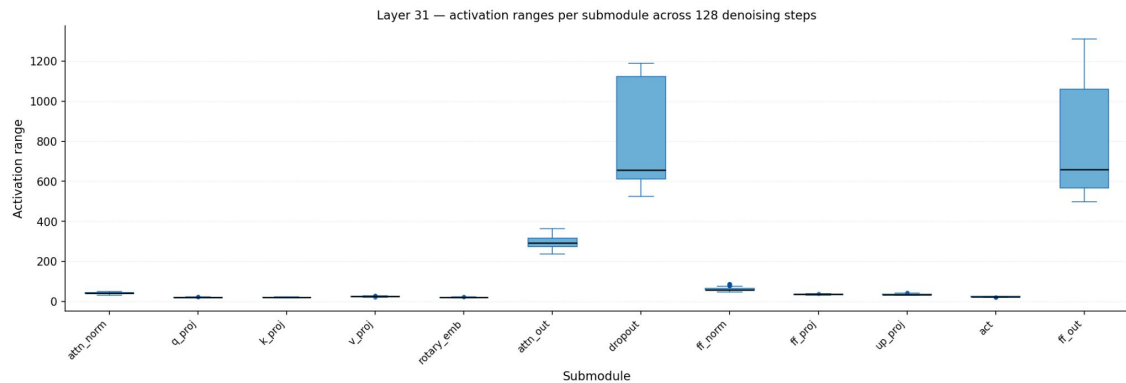
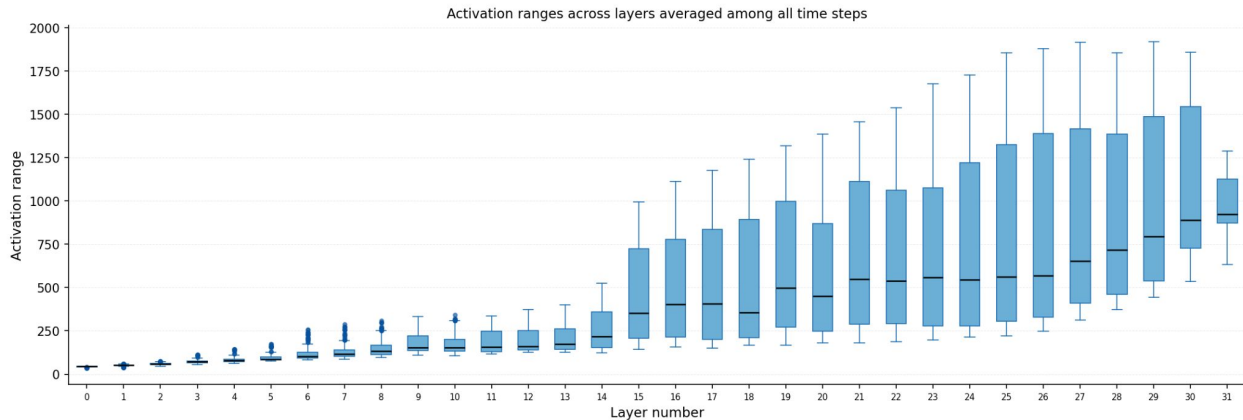


- Pearson correlation values between timesteps and layers of LLaDa (left) and Dream (right) models show strong temporal dependencies

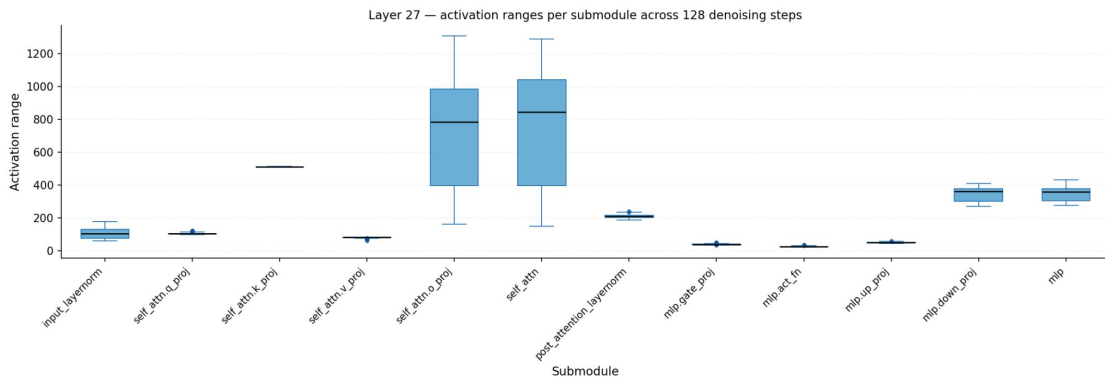
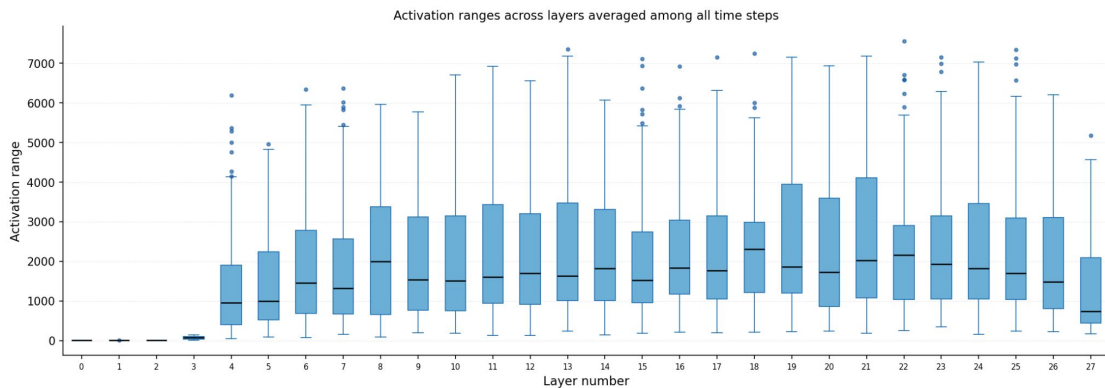
Prompt	$ r > 0.5$ (moderate)	$ r > 0.7$ (strong)
math1	84.4%	75.1%
piqa	93.8%	90.2%
hellaswag	87.1%	78.2%
winogrande	94.7%	90.7%
arc_challenge	93.8%	84.4%

Prompt	$ r > 0.5$ (moderate)	$ r > 0.7$ (strong)
math1	83.2%	73.6%
piqa	82.7%	74.6%
hellaswag	81.2%	72.1%
winogrande	86.8%	82.2%
arc_challenge	83.8%	75.1%

Variations in LLaDa Activations across Layers and Modules

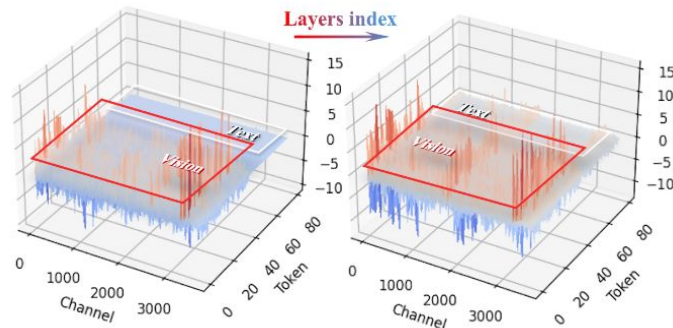
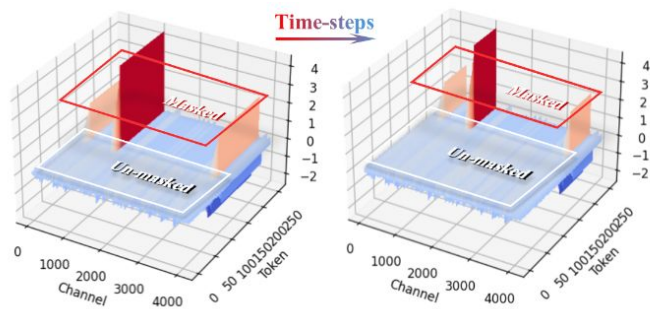


Variations in Dream Activations across Layers and Modules



FreeAct: Freeing Activations for LLM Quantization

- Optimizing transformation-based quantization methods to account for different token types in diffusion language models and multimodal LLMs.



FreeAct: Freeing Activations for LLM Quantization

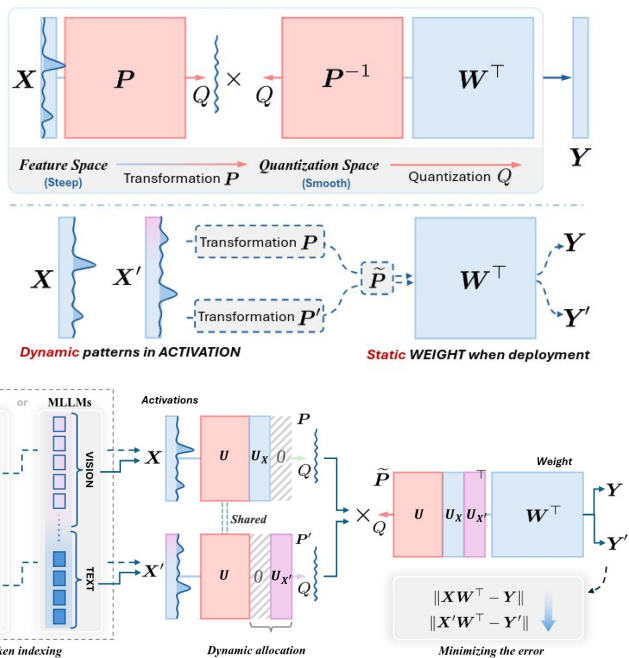


Figure 4. The overall framework of FreeAct. Different tokens are indexed to different transformation matrices according to their unique types (vision-text for MLLMs, and unmasked and masked for dLLMs). Different transformation matrices maintain a shared part together while possessing their own uniqueness, where zeros are filled in on another portion. The weight transformation matrix unites both portions to handle all the different activations. And the quantization error is minimized to optimize the quantization parameter.

Literature Review Summary - Diffusion LMs

Focus	Paper	Key Contribution	Role in the Project
DLM Architecture	Large Language Diffusion Models (Nie et al., 2025)	Proposed LLaDA diffusion language model based on the Llama architecture.	Used LLaDA-8B Base model for quantization.
DLM Architecture	Dream 7B: Diffusion Large Language Models (Ye et al., 2025)	Proposed Dream diffusion language model based on the Qwen2.5-7B architecture.	Used Dream-7B Base model for quantization.

Literature Review Summary - DLM Quantization

Focus	Paper	Key Contribution	Role in the Project
DLM Quant.	Quantization Meets dLLMs (Lin et al., 2025)	Evaluated quantization techniques on DLMs; identified massive outliers in the FF-out module of LLaDA-8B.	Motivated our activation-quantization work by surfacing challenges with W4A4.
DLM Quant.	Quant-dLLM (Zhang et al., 2025)	Weight-only extreme low-bit PTQ using masked calibration sampling and adaptive blockwise mixed precision.	Identified key DLM issues: activation distributions shift over timesteps and quantization errors accumulate.

Literature Review Summary - DLM Quantization

Focus	Paper	Key Contribution	Role in the Project
DLM Quant.	DLLMQuant (Xu et al., 2025)	Weight-activation quantization for DLMs. Claimed unmasked and low-confidence masked tokens do not propagate quantization errors, and that the post-softmax matmul in attention is the main error source.	Using the MDLM formulation we proved unmasked tokens do not propagate errors. We can use mask regions and token final scores to guide activation quantization

Literature Review Summary - Vision Diffusion Quantization

Focus		Paper	Key Contribution	Role in the Project
Vision Quant.	Diff.	Q-Diffusion (Li et al., 2023)	Addressed multistep-inference and UNet challenges via timestep-aware calibration and split quantization in shortcut layers.	Informed our box-plot analysis of how activation ranges vary across layers and timesteps.
Vision Quant.	Diff.	PTQD (He et al., 2023)	Split quantization noise into correlated and uncorrelated parts vs. the full-precision model; step-aware mixed precision selecting bit-width by per-step SNR.	Inspired our MDLM error formulation: the quantized output diverges from full precision in proportion to how far each token's logit perturbation deviates from the mean logit shift.

Literature Review Summary - Vision Diffusion Quantization

Focus	Paper	Key Contribution	Role in the Project
Vision Diff. Quant.	Temporal Dynamic Quantization (So et al., 2023)	Showed strong temporal dependency via Pearson correlation between timesteps and per-tensor activation variation; learnable TDQ network predicts the quantization interval from the timestep.	Measured the same correlation for LLaDA and Dream, confirming strong temporal dependency so a TDQ-style module may serve as a dynamic quantization scheme for DLMS.

Plan For Next Week

- Implement a dynamic activation quantization approach on DLMS based on the analysis done on activations and outliers